

GHANA COMMUNICATION TECHNOLOGY UNIVERSITY



FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

DEVELOPMENT AND IMPLEMENTATION OF A ROBOTIC APPLICATION FOR HUMAN INTENTION RECOGNITION USING MOTION AND HAND GESTURE

*A Final Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of
BSc. Computer Engineering*

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Knowledge Comes From Learning

PRESENTATION OUTLINE

01. INTRODUCTION & PROBLEM CONTEXT

- *Motivation, Problem Statement, Objectives & ERQs*

02. LITERATURE REVIEW & RESEARCH GAPS

- *Benchmark Studies (Tsitos, Mahmud, Li) & The Central Gap*

03. SYSTEM ARCHITECTURE & METHODOLOGY

- *Mechatronic Hardware, 19-D Feature MLP, LSTM, Nav2/SLAM*

04. EXPERIMENTAL VALIDATION & RESULTS

- *Latency Budget (<75 ms), Confusion Matrix & Locomotion Trials*

05. CONCLUSION & RECOMMENDATIONS

- *Summary of Contributions, ISO 15066 Compliance & Demonstration*



INTRODUCTION

- **Human-Robot Interaction (HRI)** enables robots to understand and respond to human actions.
- **Gestures and body movements** provide natural, contactless control.
- **Vision-based systems** can require high computing power..
- **Edge computing** supports real-time processing on low-power robots.
- This project combines **intention recognition, environmental awareness, and navigation.**



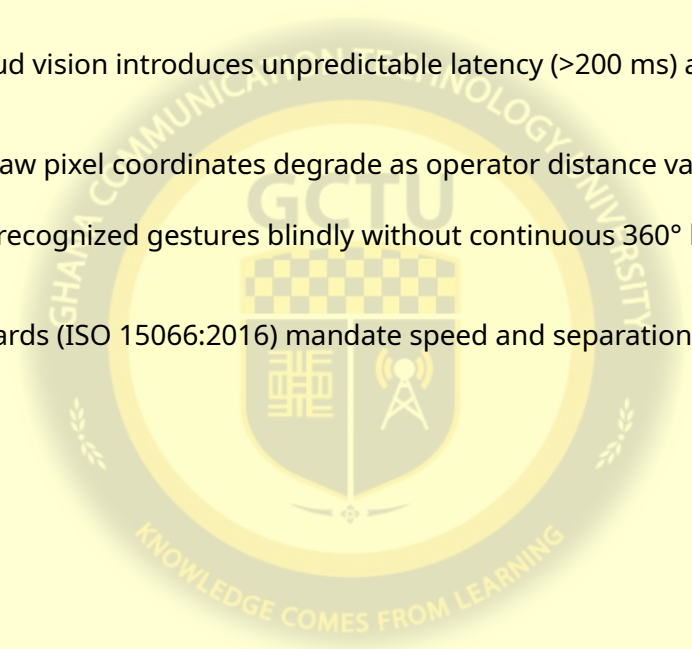
Figure 1: Autonomous Mobile Robot Performing Navigation Tasks



Figure 2: Human Interaction with an Autonomous Mobile Robot Interface

Problem Statement

- • High-Power Hardware & Cloud Latency: Offboard cloud vision introduces unpredictable latency (>200 ms) and connection drops, violating real-time robotic safety reaction limits (Tsitos et al., 2022).
- • Spatial Distance Domain Shift: Classifiers evaluating raw pixel coordinates degrade as operator distance varies (1.0m to 2.5m) due to optical perspective scaling.
- • Ungrounded Spatial Command Execution: Executing recognized gestures blindly without continuous 360° laser mapping and obstacle verification presents severe collision hazards in shared indoor workspaces.
- • Collaborative Safety Compliance: International standards (ISO 15066:2016) mandate speed and separation monitoring, requiring guaranteed deterministic halting distances (<0.45 m).



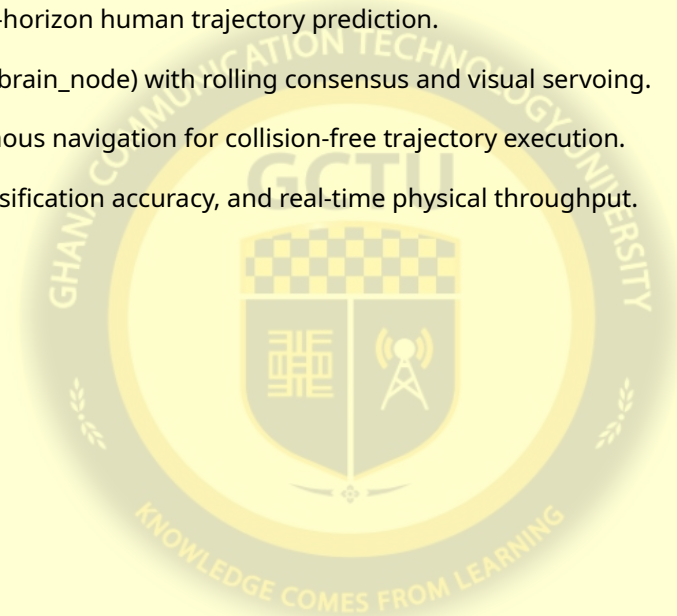
OBJECTIVES

General Objective:

- To design, implement and evaluate an edge-based real-time human intention recognition system integrating hand gestures and autonomous navigation using ROS 2.

• Specific Objectives:

- • To design and implement an edge-optimized vision pipeline using 19 scale-invariant geometric hand features.
- • To construct and balance a custom 6,000-sample gesture dataset captured directly through the onboard camera.
- • To develop an LSTM sequential recurrent model for short-horizon human trajectory prediction.
- • To implement a deterministic supervisory decision node (brain_node) with rolling consensus and visual servoing.
- • To configure and validate LiDAR SLAM and Nav2 autonomous navigation for collision-free trajectory execution.
- • To quantitatively evaluate latency budgets (<150 ms), classification accuracy, and real-time physical throughput.

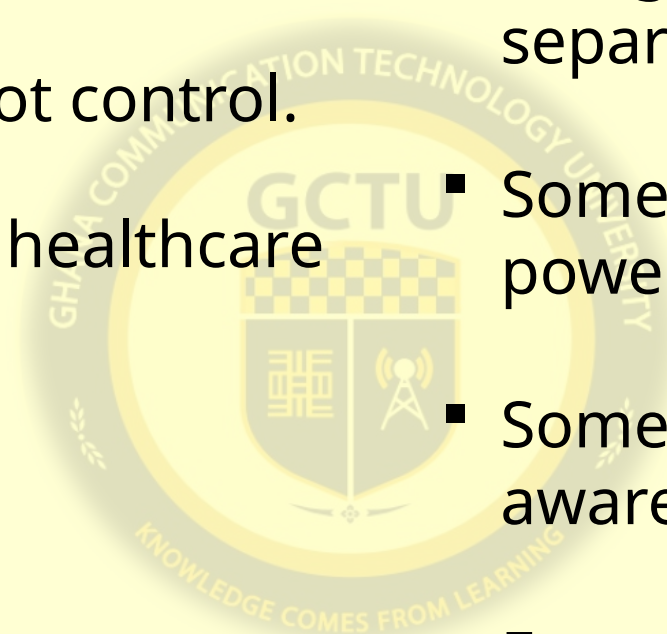


SIGNIFICANCE OF THE STUDY

- Reduces dependence on cloud computing.
- Enables contactless robot control.
- Supports industrial and healthcare applications.
- Improves human-robot interaction.
- Enables AI processing on edge devices

RESEARCH GAPS

- Existing systems often treat recognition and navigation separately.
- Some systems depend on cloud or powerful GPUs.
- Some lack environmental awareness.
- Few combine intention recognition with navigation.
- his study integrates these functions on an edge robot



LITERATURE REVIEW					
Ref / Year	Author(s)	Topic	Objectives	Methodology	Identified Gaps
[1] 2022	Tsitos et al.	Real-time Feasibility of a Human Intention Method Evaluated Through a Competitive Human-Robot Reaching Game	Predict human intention in real time using a single RGB-D camera.	OpenPose, RGB-D camera, ML models, ROS, UR3 robot.	Stationary UR3 arm without mobile base navigation or explicit gesture command channel.
[2] 2022	Mahmud et al.	3D Gesture Recognition and Adaptation for Human-Robot Interaction	Recognize 3D gestures for robot control.	Kinect, HMM, SVM, CNN, robot navigation.	Restricted to proprietary Kinect sensor; navigation evaluated only in simulation without physical latency or wheel slip.
[3] 2023	Yang et al. / Li et al.	Safe and Efficient Motion Planning for Material Transportation Robots Considering	Predict obstacle intentions for safe navigation.	Object detection, ReLU intention model. Nav2. ROS2.	Simulation only (Isaac Sim); predicts obstacle clearance but lacks touchless human supervisory control.

SYSTEM DESIGN & ENGINEERING METHODOLOGY

6-PHASE ENGINEERING RESEARCH & IMPLEMENTATION METHODOLOGY

Autonomous Mobile Robot Human Intention Recognition & Collaborative Navigation Framework

PHASE 1 Mechatronics & Hardware Integration

- Yahboom 4WD chassis & DC motor encoders
- Raspberry Pi 5 (8GB) edge compute node
- ESP32-S3 Micro-ROS real-time co-processor
- MS200 2D LiDAR (360°) & 2-DOF camera gimbal
- Decoupled dual-bus power regulation (7.4V Li-ion)

PHASE 2 Data Acquisition & Calibration

- Direct onboard camera collection (640x480)
- 6,000 balanced samples (1,000 / class)
- 6 gestures: STOP, GO, FOLLOW, LEFT, RIGHT, BACK
- Varying operational distances (1.0m to 2.5m)
- Sensor calibration under real indoor lighting

PHASE 3 Feature Engineering & Edge AI

- 21 MediaPipe 3D anatomical hand landmarks
- 19-D scale-invariant geometric feature vector
- Lightweight MLP classifier (PyTorch -> ONNX, 46KB)
- LSTM sequential motion predictor (10-frame window)
- Sub-2ms neural inference on Raspberry Pi 5 CPU

Sequential Progression to Supervisory Middleware & Navigation Integration

PHASE 4 Supervisory State Machine & Control

- Centralized ROS 2 Humble node graph (Domain 20)
- brain_node 5-frame consensus & visual servoing
- Multi-tier priority arbitration via twist_mux
- Dedicated 921,600 baud UART micro-ROS bridge
- Acoustic safety chime notification system

PHASE 5 Metric SLAM & Nav2 Navigation

- slam_toolbox 2D occupancy grid mapping (5cm)
- AMCL probabilistic particle filter localization
- Nav2 global/local costmaps with 0.25m inflation
- Extended Kalman Filter (EKF) sensor fusion
- Restamped odometry/IMU neutralizing yaw drift

PHASE 6 Empirical Benchmarking & Validation

- End-to-end component latency budget (74.2 ms)
- 180 physical closed-loop gesture locomotion trials
- 99.38% test accuracy / 96.67% physical trial accuracy
- ISO 15066 safety stopping verification (<0.36m)
- Synchronized telemetry rosbag recording & analysis



SYSTEM ARCHITECTURE

Figure 3: Hardware Architecture and Signal Distribution

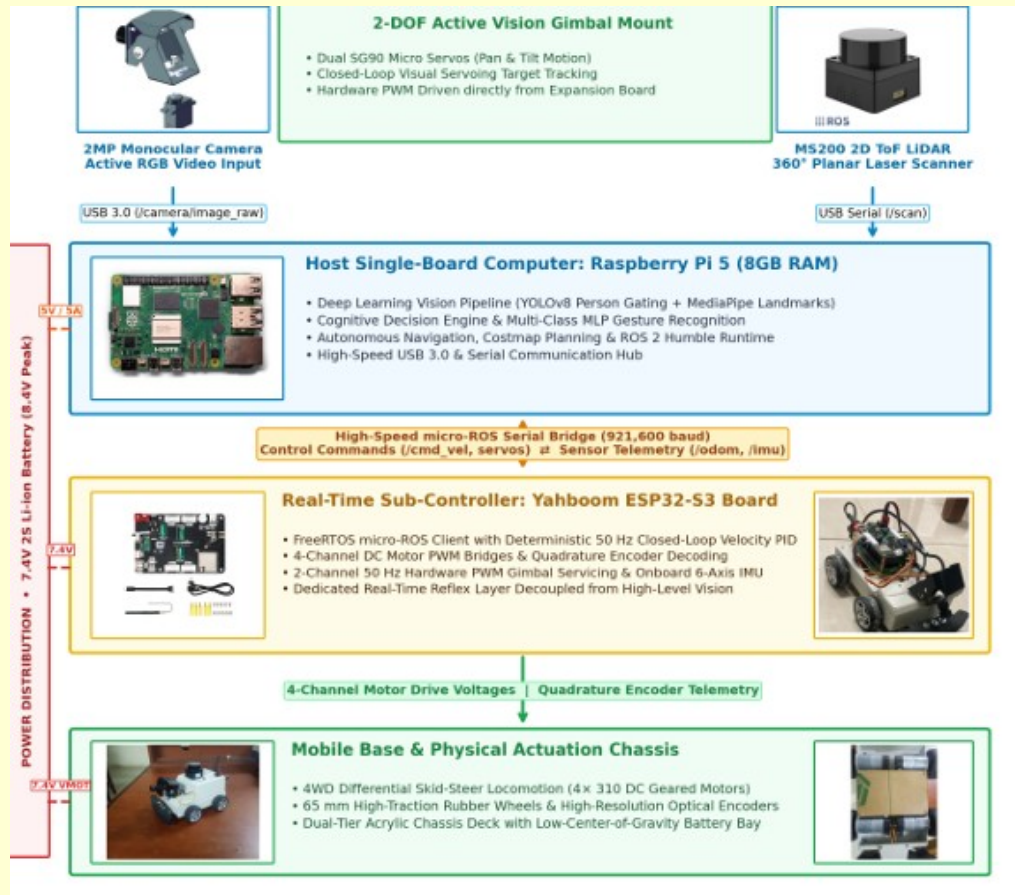
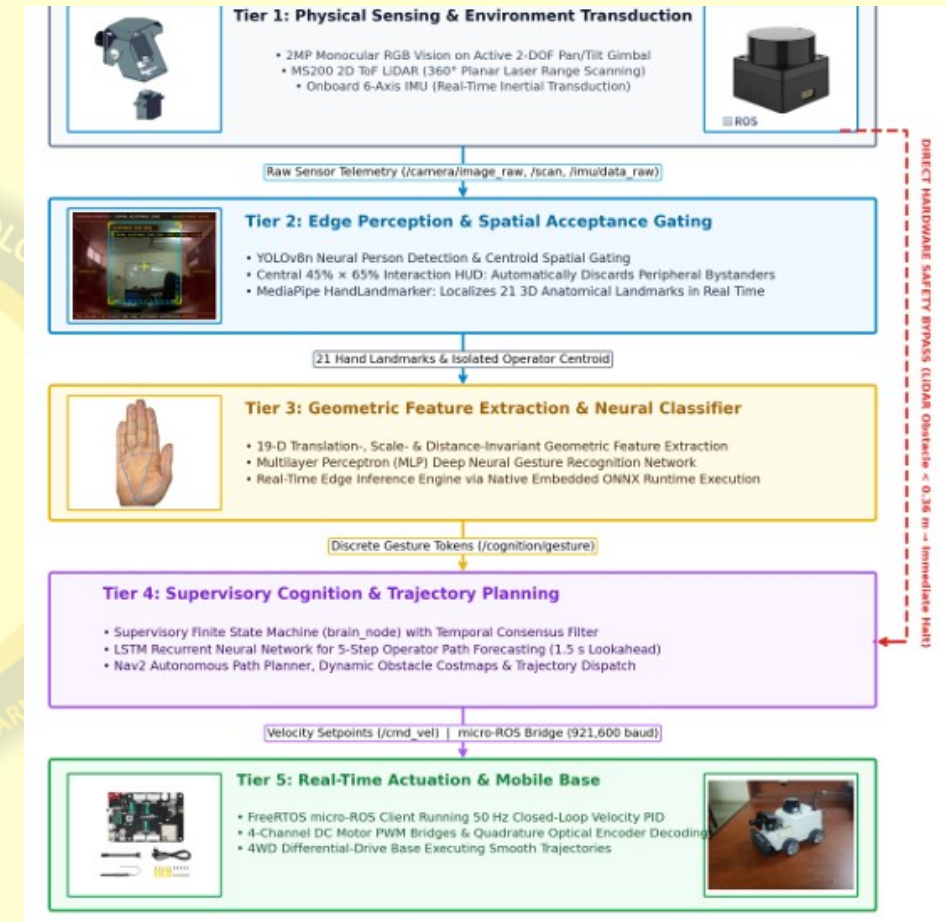


Figure 4: System Architecture



HARDWARE COMPONENTS



Figure 5: Raspberry Pi 5 (8GB RAM)

Edge SBC running ROS 2 Humble, MediaPipe, 19-D MLP, and SLAM/Nav2 directly onboard at 28.4 FPS on CPU.



Figure 6: USB Monocular Camera

Captures 640x480 RGB frames at 30 FPS for gesture classification and active gimbal tracking.



Figure 7: MS200 2D Planar LiDAR

Provides 360° laser scans at 10 Hz up to 12m for SLAM and triggers the 0.36m ISO 15066 safety halt.

HARDWARE COMPONENTS



7.4V power supply

Figure 8: Dual-Bus Power Regulation

7.4V 2S Li-ion battery pack with decoupled DC regulation isolating Pi 5 logic from motor current spikes.

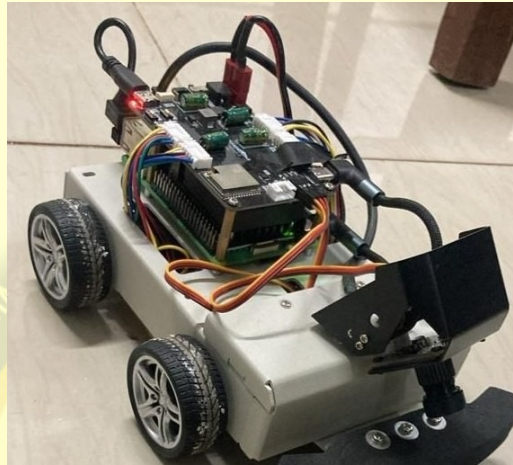


Figure 9: Yahboom 4WD Mobile Chassis

Differential-drive chassis with four geared DC motors and optical quadrature encoders.

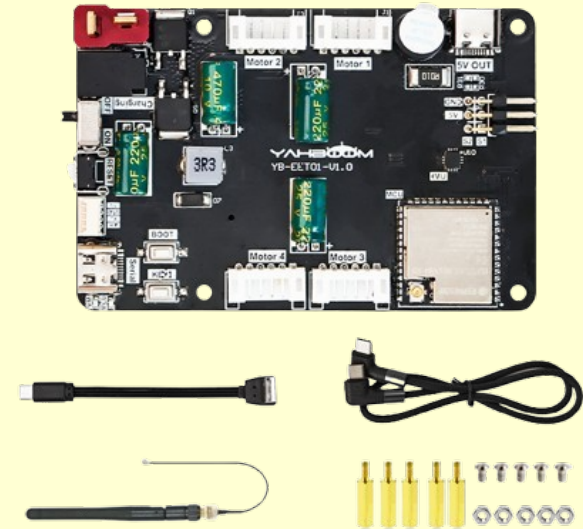


Figure 10: ESP32-S3 Micro-ROS Co-Processor

Dedicated hard real-time co-processor regulating motor PWM and reading encoders over 921,600 baud UART.

DATASET PREPARATION AND GESTURE RECOGNITION PIPELINE

- **6,000 samples** collected from the robot's own camera.
- **6 gesture classes:** Stop, Go, Left, Right, Back, Follow.
- **Balanced dataset:** 1,000 samples per gesture.
- **MediaPipe** extracts 21 hand landmarks
- Landmarks are converted into **19 normalized geometric features**
- **MLP classifier** recognizes the six gestures..

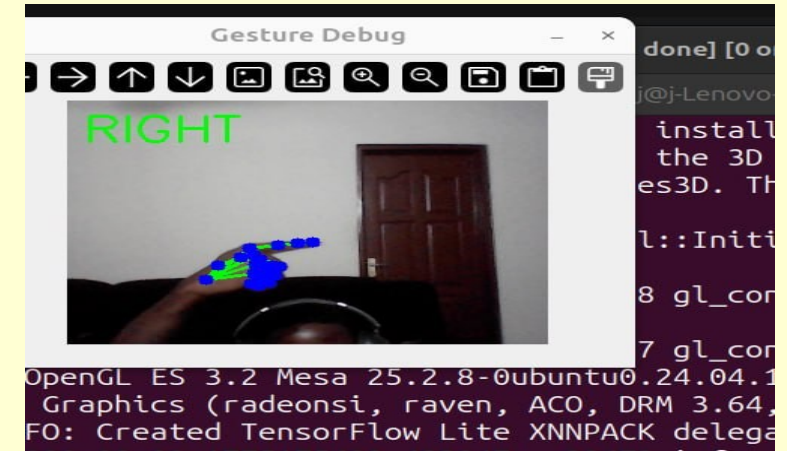


Figure 11:Live System Demonstration: The detected gesture is classified and displayed in real time.

HUMAN MOVEMENT PREDICTION (LSTM)

- Uses **MediaPipe Pose landmarks** to track human movement.
- LSTM predicts the operator's next movement direction.
- Uses **past movement patterns** to anticipate future motion.
- Helps the robot **respond proactively instead of reactively**.
- Achieved **25.8 px ADE** and **32.4 px FDE**.

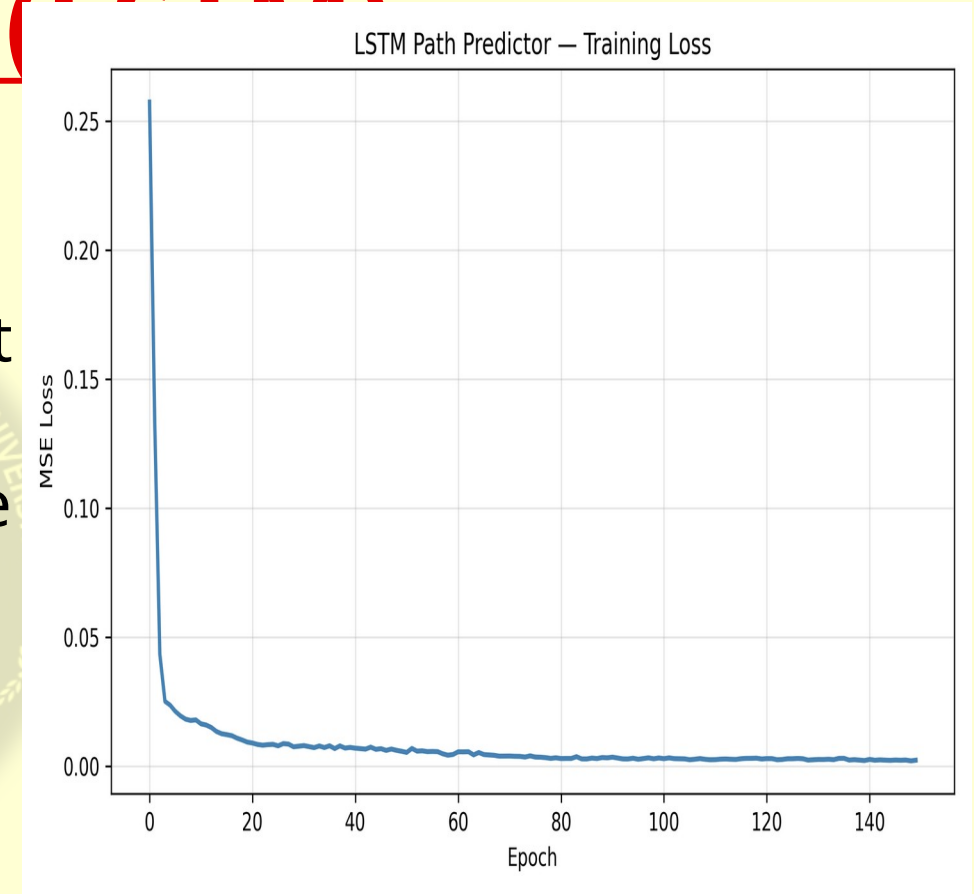


Figure 13: LSTM Path Predictor Training Loss & Convergence (ADE: 25.8 px, FDE: 32.4 px)

SLAM MAPPING AND LOCALISATION

- **SLAM Toolbox** creates a 2D map using LiDAR and wheel odometry.
- The environment is represented as an **occupancy grid**.
- **Nav2** enables point-to-point autonomous navigation.
- **AMCL** localizes the robot on the generated map.
- **Costmaps** help the robot avoid unexpected obstacles



Figure 14: Initial SLAM Run — Hourglass Rotational Drift Defect (Unsynchronized IMU Yaw)

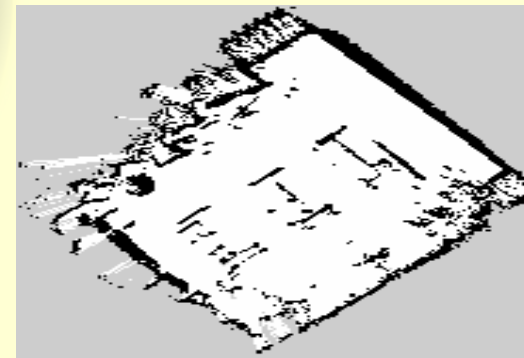


Figure 15: Final SLAM Run — Clean 5cm Metric Occupancy Grid Map (EKF Restamping Fix)

TESTING AND RESULTS

Gesture Recognition Physical Trial Results

Gesture	Total Trials	Successful Trials	Observed Accuracy (%)
Stop	30	27	90.0%
Go	30	29	96.7%
Follow	30	28	93.3%
Left	30	30	100.0%
Right	30	30	100.0%
Back	30	30	100.0%
Aggregate	180	174	96.67%

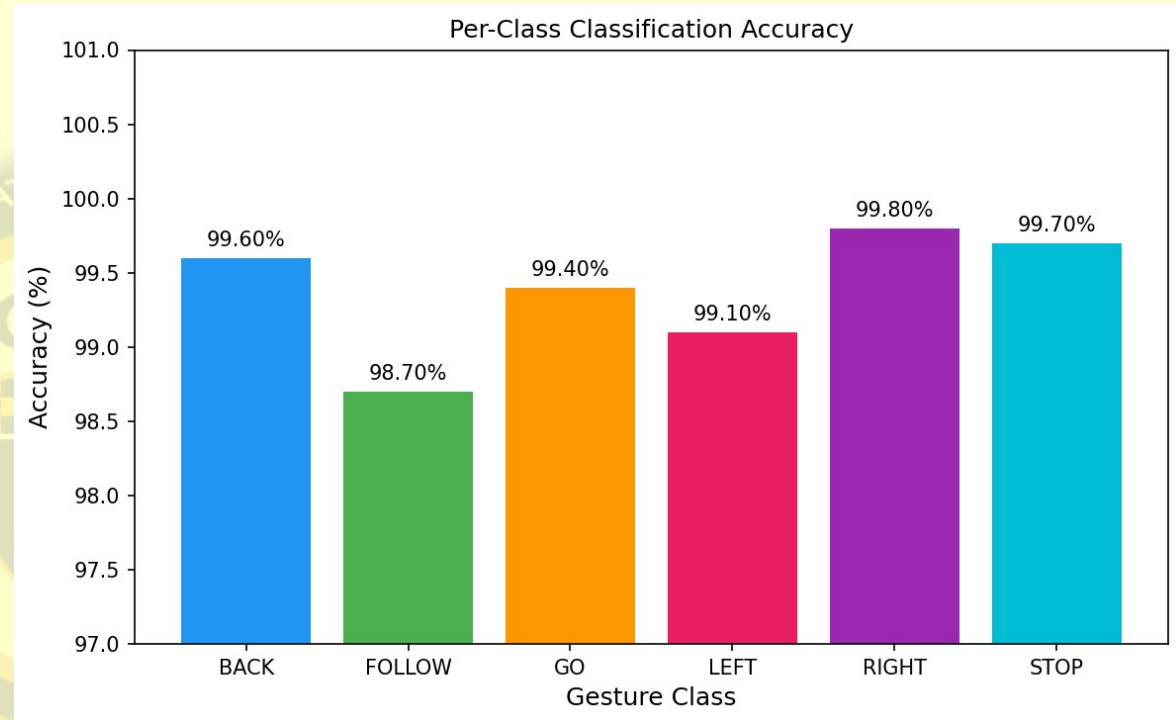


Figure 16: Per-Class Classification Accuracy of the Trained MLP Gesture Model

TESTING AND RESULTS

Movement Recognition Test Results

Movement Class	Observed Kinematic Behavior	Result
Approaching	Rapid expansion of bounding box area with stable center	Successfully Recognized
Moving Away	Continuous contraction of bounding box area with stable center	Successfully Recognized
Moving Left	Persistent leftward centroid translation across frames	Successfully Recognized
Moving Right	Persistent rightward centroid translation across frames	Successfully Recognized
Stationary	Centroid displacement within noise threshold (± 4 px)	Successfully Recognized



CONCLUSION AND FUTURE WORKS

Conclusion

- 99.38% gesture recognition accuracy.
- 96.67% physical trial accuracy.
- Successful human movement prediction.
- ROS 2 integration achieved.
- SLAM and Nav2 validated
- Real-time edge operation achieved
- Safe obstacle detection implemented

Future works

- Multi-waypoint autonomous navigation.
- Proactive human-aware navigation.
- Follow-to-Map exploration.
- NPU acceleration for AI tasks.
- Improved gimbal-chassis control
- Multi-sensor safety detection
- Larger and more diverse testing.



REFERENCES

- [1] D. Tsitos et al., "Real-Time Feasibility of a Human Intention Prediction Method Evaluated Through a Competitive Human–Robot Reaching Game," IEEE Robotics and Automation Letters, vol. 7, no. 4, pp. 10034–10041, Oct. 2022.
- [2] J. A. Mahmud, B. C. Das, J. Shin, K. M. Hasib, R. Sadik, and M. F. Mridha, "3D Gesture Recognition and Adaptation for Human–Robot Interaction," IEEE Access, vol. 10, pp. 116485–116507, 2022.
 - [3] G. Yang et al., "Safe and Efficient Motion Planning for Material Transportation Robots Considering Intention Prediction of Obstacles," in Proc. IEEE Int. Conf. Mechatronics Autom. (ICMA), Harbin, China, 2023, pp. 1179–1184.



Questions & Discussion

Thank You

- *[Transition to Live Physical Hardware Demonstration]*

